Matrix-Residual Graph Neural Networks for Graph Classification

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ABSTRACT

Background. Residual connections are widely used to stabilize graph neural networks, but residual design is often treated as a one-dimensional depth-wise choice. Multi-branch graph classifiers create a richer design space: residual information can flow along depth, across branches, or through a local two-dimensional branch-by-layer neighborhood.
Methods. We propose Matrix-Residual Graph Neural Networks as a controlled residual-topology study for graph classification. The model family compares five information-flow designs under a shared GCNConv backbone: plain message passing, vertical residual reuse, horizontal branch-wise exchange, local matrix residual reuse, and sparse/gated matrix reuse. PROTEINS, DD, and ENZYMES are evaluated with five-fold graph-level splits. We further run branch-count ablations on PROTEINS and DD, fold-0 sensitivity scans for MatrixRes and MatrixResGated, five-fold checks of selected MatrixResGated candidates, and mechanism analyses based on branch diversity, branch cosine similarity, and gradient norms.
Results. No single residual topology dominates all datasets. At the default branch count $B = 3$, HorizontalRes is strongest on PROTEINS, VerticalRes is strongest on DD, and MatrixResGated is strongest on ENZYMES. Branch-count ablations reveal a non-monotonic rise-then-fall pattern: PROTEINS tolerates moderate branch expansion, whereas DD prefers smaller branch budgets. Five-fold tuned-candidate checks show that a smaller MatrixResGated hidden dimension improves the DD result while reducing parameters; the PROTEINS sparsity candidate remains variable across folds.
Conclusions. Residual topology, branch count, and residual filtering should be selected jointly. Additional branches help only when they create useful functional diversity; otherwise, residual traffic and branch redundancy can offset the added capacity. Matrix-style reuse is therefore best understood as a tunable residual family rather than a universal replacement for simpler vertical or horizontal residual paths.

1 INTRODUCTION

Graph neural networks (GNNs) learn graph representations by repeatedly aggregating information from local neighborhoods (Kipf and Welling, 2017; Xu et al., 2019). In graph classification, the final prediction depends on whether this iterative process preserves discriminative structural and attribute information until graph-level pooling. Increasing model depth or adding parallel branches can expand capacity, but it can also introduce over-smoothing, redundant representations, unstable optimization, and unnecessary computation (Li et al., 2018; Zhao and Akoglu, 2020).

Residual connections are a common response to these difficulties. They improve gradient flow and make it possible to reuse earlier representations, and residual or dense connectivity has been transferred successfully to deeper graph convolutional networks (Li et al., 2019). However, residual connectivity is still commonly treated as a default layer-wise component rather than as the object of study. Most residual graph classifiers reuse information along depth in the same stream. Once a model contains multiple branches, this view is incomplete: residual information may also move across branches at the same depth or through diagonal branch-depth neighborhoods.

This paper studies residual reuse as a structured design problem on a branch-by-layer grid. The central question is not whether skip connections should be present, but where residual signals should flow. A grid view exposes three controlled neighborhoods: vertical reuse along depth, horizontal exchange across neighboring branches, and matrix-style reuse that combines same-branch, neighboring-branch, and diagonal residual paths. A sparse/gated matrix variant further tests whether matrix residual traffic should be filtered before injection.

The study is deliberately controlled. It fixes the graph-classification pipeline, uses the same GCNConv backbone across the main benchmark, and changes only the residual topology and residual-control mechanism. This separation is important because it keeps the empirical comparison focused on information-flow design rather than on unrelated changes in message-passing operator or readout structure.

The work is organized around three research questions:

1. **Which residual topology is strongest under a shared graph-classification protocol?** We compare plain message passing, vertical residual reuse, horizontal branch-wise exchange, local matrix residual reuse, and sparse/gated matrix reuse on PROTEINS, DD, and ENZYMES.

2. **How does branch count change residual behavior?** We test $B = 1, \ldots, 8$ for the multi-branch residual families and ask whether more branches improve accuracy monotonically or only within a useful operating region.

3. **When does controlled matrix reuse help?** We use sensitivity scans, five-fold tuned-candidate checks, and mechanism metrics to distinguish promising residual-control settings from fold-specific artifacts.

The main contributions are:

- **A branch-by-layer residual formulation for graph classification.** Vertical, horizontal, matrix, and sparse/gated matrix residual families are expressed as local neighborhoods on the same grid, making their differences explicit while preserving a common backbone and classifier.

- **A controlled five-fold benchmark and branch-count ablation.** The experiments show that residual topology is dataset-dependent: HorizontalRes is strongest on PROTEINS at $B = 3$, VerticalRes is strongest on DD, and MatrixResGated is strongest on ENZYMES, while branch-count curves follow a rise-then-fall pattern.

- **Evidence that branch expansion is not merely a capacity knob.** Branch diversity, branch cosine similarity, and gradient norms show that additional branches help only when they create useful separation without excessive residual traffic or optimization weakening.

- **A conservative validation of MatrixResGated tuning.** Fold-0 sensitivity scans identify candidate operating regions, and five-fold reruns show that the smaller DD MatrixResGated setting is the most stable tuned improvement, whereas the PROTEINS sparsity candidate is not yet a robust full-fold gain.

2 MATERIALS AND METHODS

This section defines the graph-classification task, the Matrix-Residual architecture family, the benchmark protocol, and the mechanism metrics. The guiding principle is to isolate residual topology while holding the remaining learning pipeline fixed.

2.1 Problem definition and notation

Let $\mathscr{G} = (\mathscr{V}, \mathscr{E})$ be a graph with node feature matrix $\mathbf{X} \in \mathbb{R}^{n \times d_{\text{in}}}$ and adjacency structure $\mathbf{A}$. For a dataset $\mathscr{D} = \{(\mathscr{G}_i, y_i)\}_{i=1}^{N}$, the objective is to learn $f_\theta : \mathscr{G} \to \hat{y}$, where $\hat{y}$ is the predicted graph label.

The compared model families share the same graph-classification wrapper:

- **Plain**: stacked message passing without explicit residual reuse.

- **VerticalRes**: same-branch residual reuse from the previous layer.

- **HorizontalRes**: cross-branch exchange between adjacent branches at the same depth.

- **MatrixRes**: local two-dimensional reuse over vertical, horizontal, and diagonal neighbors.
- **MatrixResGated**: matrix reuse with gated or sparse residual filtering.

2.2 Shared graph-classification pipeline

Let B be the number of branches and L be the number of message-passing layers. The hidden state at branch b and layer ℓ is written as $\mathbf{H}_b^{(\ell)}$, where $b \in \{1, \ldots, B\}$. A branch-local propagation step is

$$\mathbf{Z}_b^{(\ell)} = \phi_b^{(\ell)} \left(\mathbf{H}_b^{(\ell-1)}, \mathbf{A} \right), \tag{1}$$

where ϕ is instantiated as GCNConv in the main benchmark. The residual-enhanced update is

$$\mathbf{H}_b^{(\ell)} = \mathbf{Z}_b^{(\ell)} + \sum_{(b',\ell') \in \mathscr{N}(b,\ell)} \mathscr{R}_{(b',\ell') \to (b,\ell)} \left(\mathbf{H}_{b'}^{(\ell')} \right). \tag{2}$$

The only term that changes across architectures is the residual neighborhood $\mathscr{N}(b,\ell)$ and the residual transform $\mathscr{R}$.

After the final message-passing layer, each branch is pooled to a graph-level vector and the branch vectors are combined before classification. The classifier is trained with cross-entropy loss. This keeps the comparison focused on residual topology rather than on architecture-specific readout heads.

2.3 Residual neighborhoods

Plain. The plain baseline has no residual neighborhood:

$$\mathscr{N}(b,\ell) = \emptyset. \tag{3}$$

VerticalRes. Vertical residual reuse injects the previous layer on the same branch:

$$\mathscr{N}(b,\ell) = \{(b,\ell-1)\}. \tag{4}$$

HorizontalRes. Horizontal residual reuse exchanges information across adjacent branches at the same depth:

$$\mathscr{N}(b,\ell) = \{(b-1,\ell),(b+1,\ell)\}, \tag{5}$$

with boundary branches using only valid neighbors.

MatrixRes. Matrix residual reuse combines vertical, horizontal, and diagonal neighborhoods:

$$\mathscr{N}(b,\ell) = \{(b,\ell-1),(b-1,\ell),(b+1,\ell),(b-1,\ell-1),(b+1,\ell-1)\}. \tag{6}$$

This local stencil is intentionally not dense all-to-all mixing. It preserves an interpretable residual structure over the branch-by-layer grid.

MatrixResGated. MatrixResGated uses the same neighborhood as MatrixRes, but the residual transform controls the injected signal:

$$\mathscr{R}(\mathbf{U}) = \begin{cases} \alpha \mathbf{U}, & \text{gated residual injection,} \\ \operatorname{softshrink}_\lambda(\mathbf{U}), & \text{sparse residual filtering.} \end{cases} \tag{7}$$

This variant tests whether matrix residual reuse should be passed densely or selectively.

2.4 Datasets and evaluation protocol

The main benchmark uses PROTEINS, DD, and ENZYMES from TUDataset (Morris et al., 2020). PROTEINS and DD are binary protein-structure graph classification benchmarks, while ENZYMES is a six-class enzyme classification benchmark. All experiments use the raw benchmark features and graph structures without task-specific feature engineering.

All main results use graph-level five-fold evaluation. For fold k, the test accuracy is

$$\mathrm{Acc}_k = \frac{1}{|\mathscr{T}_k|} \sum_{(\mathscr{G}_i, y_i) \in \mathscr{T}_k} \mathbf{1}[\hat{y}_i = y_i]. \tag{8}$$

Across folds, we report

$$\overline{\mathrm{Acc}} = \frac{1}{5}\sum_{k=1}^{5} \mathrm{Acc}_k, \tag{9}$$

and the fold-level standard deviation.

2.5 Training configuration

All main benchmark runs use a shared GCNConv backbone with default branch count $B = 3$. The hidden dimension is 64 unless a sensitivity setting changes it, and all model families use the same optimizer, early-stopping logic, graph pooling, and fold protocol. The branch-count ablation evaluates $B = 1, \ldots, 8$ for HorizontalRes, MatrixRes, and MatrixResGated on PROTEINS and DD. The sensitivity scan evaluates MatrixRes and MatrixResGated on fold 0 at $B = 3$, varying learning rate, dropout, hidden dimension, sparsity strength, and gate initialization. Selected MatrixResGated candidates are then rerun across all five folds.

Table 1. Key experimental settings shared by the main benchmark.

Parameter	Value
Message-passing operator	GCNConv
Default branch count	3
Default hidden dimension	64
Evaluation	Five-fold graph classification
Branch-count ablation	$B = 1, \ldots, 8$
Primary metric	Mean best test accuracy
Sensitivity scope	Fold 0, $B = 3$

2.6 Mechanism metrics

To interpret branch-count behavior, we track three mechanism indicators. Branch diversity measures the mean pairwise distance between branch representations. Branch cosine similarity measures whether branches remain redundant. Mean gradient norm tracks optimization weakening as branch count and residual traffic grow. CKA and residual-traffic summaries are retained as supplementary diagnostics.

3 RESULTS

All results are reported as five-fold mean best test accuracy $\pm$ fold-level standard deviation unless otherwise stated.

3.1 Overall benchmark performance

Table 2 reports the default $B = 3$ comparison. Residual topology changes the ranking, but the ranking is dataset-dependent. On PROTEINS, HorizontalRes gives the strongest mean accuracy at 0.7062 ± 0.0323. On DD, VerticalRes performs best at 0.7249 ± 0.0247. On ENZYMES, MatrixResGated obtains the best mean accuracy at 0.2933 ± 0.0470, although the low absolute accuracy and higher variance make this dataset better interpreted as a difficult supporting benchmark.

Table 2. Main benchmark results with GCNConv and default branch count $B = 3$.

Dataset	Plain	VerticalRes	HorizontalRes	MatrixRes	MatrixResGated
PROTEINS	0.7044 ± 0.0387	0.6981 ± 0.0402	$\mathbf{0.7062 \pm 0.0323}$	0.7035 ± 0.0247	0.7026 ± 0.0238
DD	0.7053 ± 0.0517	$\mathbf{0.7249 \pm 0.0247}$	0.7113 ± 0.0409	0.7122 ± 0.0362	0.7181 ± 0.0259
ENZYMES	0.2683 ± 0.0343	0.2550 ± 0.0455	0.2633 ± 0.0584	0.2750 ± 0.0577	$\mathbf{0.2933 \pm 0.0470}$

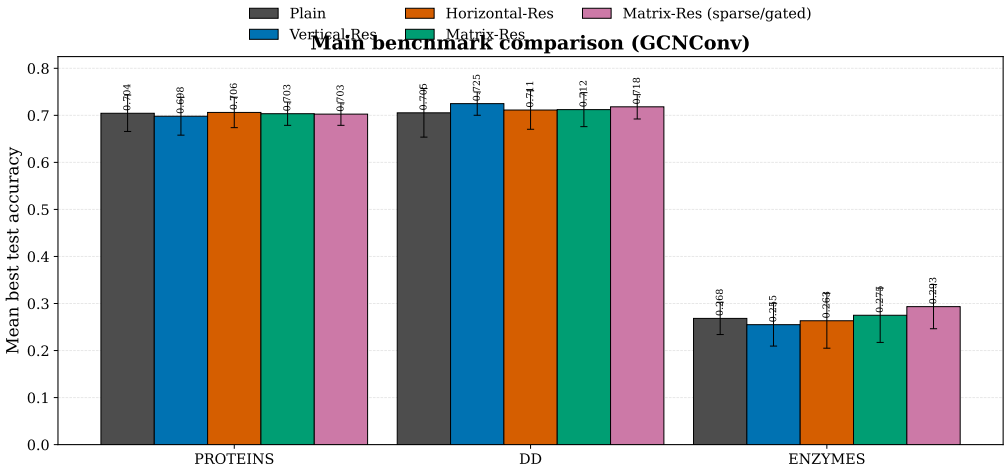

Figure 1. Main GCNConv benchmark at $B = 3$. Bars show five-fold mean best test accuracy with fold-level standard-deviation error bars.

Figure 1 makes the split regime visually explicit. The default setting does not support a universal residual winner; instead, each dataset favors a different residual neighborhood.

3.2 Branch-count ablation

Branch count has a non-monotonic effect. Table 3 reports the best branch count for each topology in the ablation. On PROTEINS, HorizontalRes peaks at $B = 4$, MatrixRes peaks at $B = 6$, and MatrixResGated peaks at $B = 3$. On DD, the preferred branch budget is smaller: HorizontalRes and MatrixRes peak at $B = 2$, while MatrixResGated is strongest at $B = 1$.

Table 3. Best branch-count setting for each topology in the ablation.

Dataset	Model	Best B	Accuracy
PROTEINS	HorizontalRes	4	0.7125 ± 0.0286
PROTEINS	MatrixRes	6	0.7062 ± 0.0210
PROTEINS	MatrixResGated	3	0.7098 ± 0.0219
DD	HorizontalRes	2	0.7275 ± 0.0304
DD	MatrixRes	2	0.7206 ± 0.0411
DD	MatrixResGated	1	0.7283 ± 0.0244

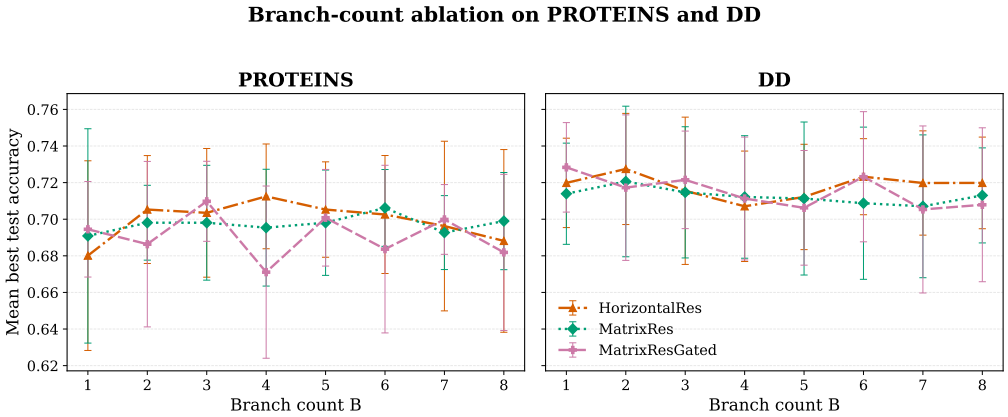

Figure 2. Branch-count ablation for HorizontalRes, MatrixRes, and MatrixResGated on PROTEINS and DD. Accuracy improves only within a useful operating region and then saturates or declines.

The ablation shows that branch count should not be read as a simple capacity knob. Increasing B changes the residual graph itself: it adds branches, residual paths, residual traffic, and optimization interactions. PROTEINS tolerates a moderate branch budget, while DD is more sensitive to redundant branch expansion.

3.3 MatrixResGated sensitivity

The fold-0 sensitivity scan shows that MatrixResGated is tunable, but its preferred region differs between datasets. On PROTEINS, the baseline MatrixResGated configuration reaches 0.6726, and mild sparsification with $\lambda = 0.02$ improves fold-0 accuracy to 0.6951. Stronger sparsification with $\lambda = 0.1$ reduces accuracy to 0.6278. On DD, the baseline fold-0 score is 0.7300, while $lr = 0.001$, $dim = 32$, and $gate_init = 0.2$ each reach 0.7511.

Table 4. Strongest MatrixResGated fold-0 sensitivity settings.

Dataset	Setting	Accuracy	Interpretation
PROTEINS	baseline	0.6726	default controlled matrix reuse
PROTEINS	$\lambda = 0.02$	0.6951	mild sparsification improves fold 0
PROTEINS	$\lambda = 0.1$	0.6278	strong sparsification is harmful
DD	baseline	0.7300	default controlled matrix reuse
DD	$lr = 0.001$	0.7511	conservative optimization helps
DD	$dim = 32$	0.7511	smaller hidden dimension helps
DD	$gate_init = 0.2$	0.7511	lower initial gate helps

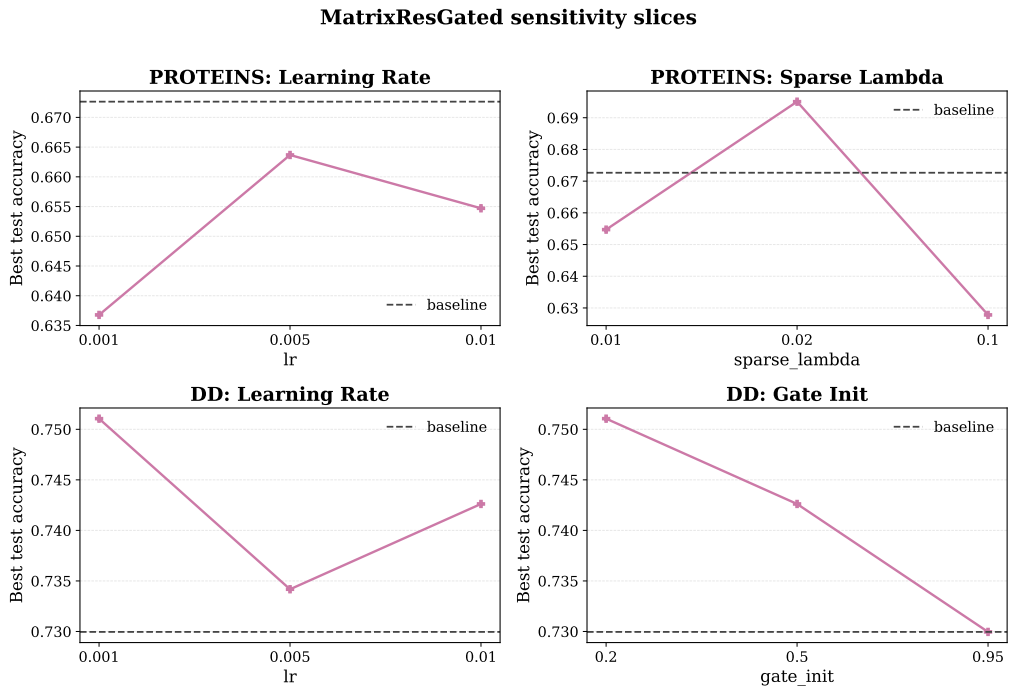

Figure 3. Fold-0 MatrixResGated sensitivity scan at $B = 3$. The scan identifies candidate operating regions but is not treated as final hyperparameter evidence.

3.4 Five-fold check of tuned MatrixResGated candidates

To distinguish candidate discovery from evaluation, selected MatrixResGated settings were rerun across all five folds. Table 5 reports these follow-up checks. On DD, the smaller hidden dimension is the strongest tuned candidate, reaching 0.7215 ± 0.0330 while reducing parameters from 42,371 to 15,043. The lower learning rate is essentially tied with the default MatrixResGated benchmark, and the lower gate initialization is weaker. On PROTEINS, $\lambda = 0.02$ reaches 0.6909 ± 0.0447, below the default

MatrixResGated result, showing that the fold-0 sparsity gain is not stable across all folds in the current run.

Table 5. Five-fold checks of selected MatrixResGated candidates.

Candidate	Dataset	Accuracy	Mean loss	Parameters
`PROTEINS_sparse_lambda_0.02`	PROTEINS	0.6909 ± 0.0447	0.6104	38,339
`DD_lr_0.001`	DD	0.7181 ± 0.0336	0.5737	42,371
`DD_dim_32`	DD	$\mathbf{0.7215 \pm 0.0330}$	0.5717	15,043
`DD_gate_init_0.2`	DD	0.7131 ± 0.0213	0.5820	42,371

3.5 Mechanism analysis

The mechanism summaries support a common explanation for the rise-then-fall trend. Increasing branch count initially creates useful diversity: branches become less identical and can cover complementary feature subspaces. Past the useful regime, diversity alone is not sufficient. Residual traffic grows quickly, branch similarity may remain high, and gradient norms can weaken.

For HorizontalRes on PROTEINS, moving from $B = 1$ to $B = 4$ increases branch diversity from 0.0000 to 1.1450, while branch cosine decreases from 1.0000 to 0.8417. This corresponds to the accuracy improvement. Beyond $B = 4$, diversity continues increasing, reaching 1.5441 at $B = 8$, but accuracy declines to 0.6881. MatrixRes shows a similar trend at a larger branch budget, peaking at $B = 6$. DD is more conservative: branch expansion beyond $B = 2$ often adds residual complexity without enough functional separation.

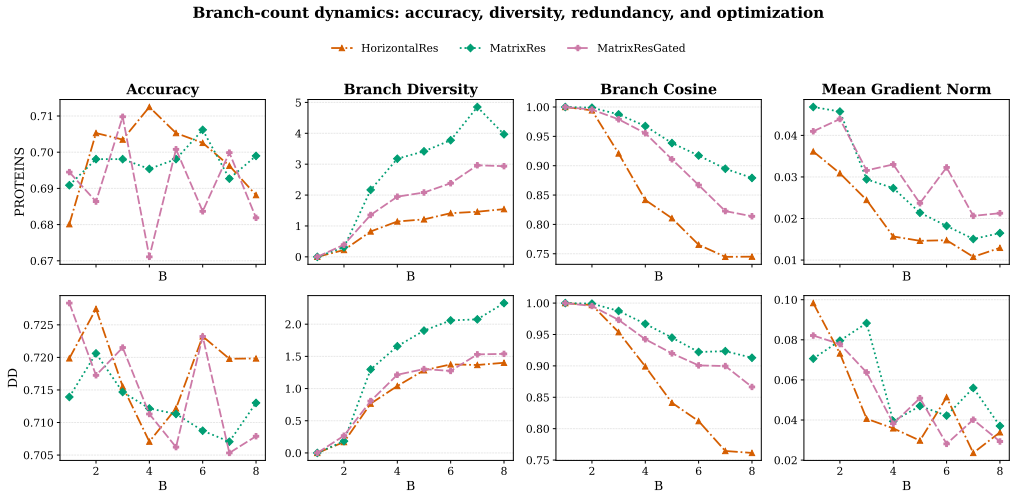

Figure 4. Mechanism summary linking branch-count accuracy to branch diversity, branch cosine similarity, and mean gradient norm.

4 DISCUSSION

This section answers the three research questions in light of the benchmark, ablation, tuned-candidate, and mechanism evidence.

4.1 RQ1: Which residual topology is strongest?

The main benchmark argues against a universal residual topology. At $B = 3$, HorizontalRes is strongest on PROTEINS, VerticalRes is strongest on DD, and MatrixResGated is strongest on ENZYMES. These outcomes indicate that the useful residual neighborhood depends on dataset-specific tolerance for branch specialization, residual traffic, and optimization complexity.

Answer: Residual topology is dataset-dependent. A graph-classification benchmark should compare vertical, horizontal, and matrix-style reuse rather than assuming that one residual direction is universally best.

4.2 RQ2: How does branch count change residual behavior?

The branch-count ablation shows that B is not only a capacity parameter. Increasing B adds branches, residual paths, and optimization interactions. On PROTEINS, moderate expansion is useful: HorizontalRes peaks at $B = 4$, MatrixRes at $B = 6$, and MatrixResGated at $B = 3$. On DD, smaller budgets are preferable, with peaks at $B = 1$ or $B = 2$. The mechanism summaries explain this split. Branch diversity may increase with B, but if branch cosine remains high or gradients weaken, the extra branches do not translate into better classification.

Answer: Branch count helps when it creates useful functional diversity, and hurts when residual complexity grows faster than branch separation or optimization quality.

4.3 RQ3: When does controlled matrix reuse help?

MatrixResGated defines a tunable family rather than a plug-and-play winner. The fold-0 sensitivity scan suggests that PROTEINS can benefit from mild sparsification, while DD benefits from conservative optimization, smaller hidden dimension, and lower initial gate. The five-fold candidate reruns refine this interpretation. On DD, the smaller hidden dimension remains beneficial and parameter-efficient. On PROTEINS, the mild-sparsity candidate does not preserve its fold-0 advantage across all folds.

Answer: Controlled matrix reuse is useful, but candidate settings must be validated across folds. The current evidence supports a smaller controlled matrix model on DD and leaves PROTEINS sparsification as a promising but not yet stable direction.

4.4 Limitations

Several limitations bound the generality of these findings. First, the main benchmark uses GCNConv only. This makes the comparison controlled but leaves open whether the same residual-topology trends hold for GINConv, GraphSAGE, or attention-based operators. Second, the sensitivity scan is fold-0 only; although selected candidates were rerun across five folds, that follow-up is limited to MatrixResGated on PROTEINS and DD. Third, ENZYMES has low absolute accuracy and high variability, so its result should be interpreted cautiously. Finally, the mechanism analysis is correlational. Branch diversity, cosine similarity, and gradient norms support a coherent explanation, but they do not prove causality.

4.5 Future directions

Future work should extend the comparison to additional message-passing operators, larger graph-classification datasets, and adaptive residual-neighborhood selection. A natural next step is to replace manually chosen vertical, horizontal, or matrix stencils with a learned local residual policy. The tuned-candidate validation should also be broadened beyond the current MatrixResGated checks to determine whether dataset-specific residual control can be selected reliably.

5 CONCLUSIONS

This paper studies residual connectivity for graph classification as a structured branch-by-layer design problem. Under a shared GCNConv benchmark, residual topology matters, but its value depends on the dataset and branch budget. HorizontalRes is strongest on PROTEINS at the default branch count, VerticalRes is strongest on DD, and MatrixResGated is strongest on ENZYMES. The branch-count ablations further show that adding branches is useful only within a limited operating region.

The mechanism analysis explains why branch expansion is not automatically beneficial. Additional branches can increase representational diversity, but they also increase residual traffic and may preserve redundant branch similarity or weaken optimization. Matrix-style reuse is therefore best viewed as a controlled residual family whose effectiveness depends on the branch budget and residual filtering strategy.

The practical implication is that residual topology, branch count, and residual control should be evaluated jointly. Before treating a sensitivity result as a performance claim, candidate settings should be rerun across folds. In the current experiments, this validation supports a smaller MatrixResGated model on DD, while the PROTEINS sparsity candidate remains less stable.

6 ACKNOWLEDGMENTS

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A APPENDIX

A.1 Result sources

The manuscript uses the following summary files as the source of truth:

- `records/LATEST/summaries/benchmark_summary.csv`
- `records/LATEST/summaries/branch_ablation_summary.csv`
- `records/LATEST/summaries/parameter_sensitivity_summary.csv`
- `records/LATEST/summaries/tuned_candidate_summary.csv`
- `records/LATEST/summaries/mechanism_compact_summary.csv`

A.2 Supplementary diagnostics

Additional fold-level rows, CKA summaries, residual traffic summaries, and full sensitivity tables are retained in the repository records directory for supplementary conversion.

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